

# PROJECT PROPOSAL

## Title: Smart Billing and Inventory Management System

Course: OOP

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### 1. Project Overview

This project is a C++ console-based Smart Billing and Inventory Management System designed to simplify retail and billing operations. The system allows an administrator to manage products, generate invoices, and apply discounts while maintaining accurate records through file handling.

The application demonstrates key Programming Fundamentals (PF) concepts such as loops, arrays, functions, file handling, strings, and conditionals, offering a practical implementation of a mini business management tool.

In the next development phase, the project will be extended to Object-Oriented Programming (OOP) using classes such as Inventory, Item, Customer, and Invoice.

### 2. Objectives

- To create a functional and interactive billing and inventory management system using C++.
- To implement Programming Fundamentals concepts such as arrays, loops, functions, and file handling.
- To ensure data persistence using text file storage for inventory and invoices.
- To design a secure and user-friendly system using password authentication.
- To make the project scalable for OOP conversion in later stages.

### 3. System Functionalities

| Feature                     | Description                                                           | Function(s)    |
|-----------------------------|-----------------------------------------------------------------------|----------------|
| <b>Login Authentication</b> | Secure system access with password protection and limited attempts.   | login()        |
| <b>Add Inventory</b>        | Allows the admin to add new products (name, quantity, price) into the | addInventory() |

|                                |                                                                                    |                                       |
|--------------------------------|------------------------------------------------------------------------------------|---------------------------------------|
|                                | system.                                                                            |                                       |
| <b>View Inventory</b>          | Displays all current items in stock from the file.                                 | viewInventory()                       |
| <b>Edit Inventory</b>          | Enables updating an existing item's quantity or price.                             | editInventory()                       |
| <b>Delete Inventory</b>        | Deletes specific items after user confirmation.                                    | deleteInventory()                     |
| <b>Generate Invoice</b>        | Creates detailed customer invoices, updates stock, and applies optional discounts. | generateInvoice(),<br>applyDiscount() |
| <b>View Invoices</b>           | Displays all saved invoices with purchase history.                                 | viewInvoices()                        |
| <b>Date &amp; Time Display</b> | Shows the current system date and time for billing records.                        | printDateTime()                       |
| <b>Screen Clear Function</b>   | <b>Clears the console for a clean interface.</b>                                   | <b>clearScreen()</b>                  |

#### 4. Programming Concepts Used

- Functions: Used for modularity; each task (add, edit, delete, etc.) is handled separately.
- Loops: Used for menus, repeated inputs, and data validation.
- Arrays & Strings: Store product names, customer details, and invoice data.
- Conditionals: Validate user input, apply discounts, and control access.
- File Handling: Used to store and retrieve inventory data and invoice history from .txt files.
- Input/Output Streams: Used for interactive console operations.
- Data Validation: Ensures correct numeric and text inputs.

#### 5. Future OOP Mapping

| Class Name | Attributes                                | Functions / Responsibilities                |
|------------|-------------------------------------------|---------------------------------------------|
| User       | username, password                        | login, authentication                       |
| Item       | name, quantity, price                     | addItem, updateItem, deleteItem             |
| Inventory  | itemList, filePath                        | viewInventory, searchItem, saveToFile       |
| Customer   | name, phone, invoiceList                  | inputDetails, validateContact               |
| Invoice    | customer, purchasedItems, total, discount | generateInvoice, applyDiscount, saveInvoice |

#### 6. Expected Outcomes

- A functional and interactive billing system for small businesses.
- Hands-on implementation of C++ fundamentals in a real-world scenario.
- Scalable code design ready for object-oriented extension.
- Improved understanding of file I/O, data validation, and modular programming.